Cristiano Cavo

CURRICULUM VITAE · SOFTWARE ENGINEER · MANAGER · AGENTIC SYSTEMS

Manager at PricewaterhouseCoopers and chartered Information Engineer. At PwC I lead the development of a **blockchain banking platform** in production and the onboarding of the banks adopting it. Alongside, I build **agentic systems**, mostly with Claude: long-term memory for agents, orchestration of agent teams, skills and verifiable knowledge bases.

OPEN TO PROJECT-BASED COLLABORATIONS · FIXED FEE · SELF-MANAGED SCHEDULE

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GitHub strawberry-code LinkedIn cristiano-cavo Based in Gavi (AL) · Piedmont

WHAT I OFFER

- **Agentic systems I use every day** · long-term memory for agents, orchestration of agent teams, skills, RAG and MCP, built for the real work of my team and myself
- **Verifiable knowledge for LLMs** · knowledge graphs and second brains where every claim points back to its source, with embeddings and multi-user sync
- **6+ years of enterprise delivery** · high-stakes banking, government and fintech, systems handling more than 300,000 logins a day
- **End-to-end solution architecture** · from design to client integration, including the mediation work between parties

HOW I WORK WITH YOU

- **Project-based engagements** · you tell me what you need, we agree on goal and deadline (say three months) and a fixed fee
- **Self-managed schedule** · no 9-to-6 hours and no imposed weekly planning, what counts is the result delivered by the agreed date
- **Compatible with my role at PwC** · work outside the perimeter of its clients, with limited and agreed availability
- **Topics I push on** · research and development with agents, Claude first but not only: memory, orchestration, skills, knowledge-work automation
- **Applied research too** · prototypes, feasibility studies, literature reviews run with agents and systematic source checking
- **To get started** · two lines about the problem to cristianocavo@icloud.com are enough

PROFESSIONAL PATH

Oct 2025 — today Milan · Italy

Manager — PricewaterhouseCoopers

I keep leading the programme for a **technology innovation consortium of the banking sector**: a permissioned blockchain platform that digitises salary-backed loans between a public body and financial institutions, now in production with the first banks. I handle **bank integration and support**, software management (builds, releases, quality), functional analysis and development. Alongside, I bring **agentic AI** into the team's work: agent orchestration, shared memory, skills. Technical lead, mentoring, client interface.

Agentic AI · Agent orchestration · LLM / RAG / MCP · Delivery Lead · Bank onboarding · Functional analysis · Corda · Kotlin · Fullstack

Feb 2022 — Oct 2025 Milan · Italy

Senior Associate — PricewaterhouseCoopers

Three years on a programme for a **technology innovation consortium of the banking sector**: blockchain-based digitisation of financial instruments between banks and public bodies. I designed the platform **from scratch**, from architecture to functional and software design, and managed the direct client relationship and the mediation between the parties involved. Development on **Corda R3** (v4 and v5), Kotlin, Python, Cloud SaaS. From 2024 I started a parallel focus on prompt engineering and AI agents, joining the GenAI practice.

Banking R&D Consortium · Solution Architect · Stakeholder Mgmt · Finance / Government · Corda 5 · Kotlin · Python · Cloud SaaS · Prompt Engineer

Dec 2019 — Feb 2022 Milan · Italy

Associate — PricewaterhouseCoopers

Two years dedicated to one of the **largest Italian and European banks** for **employee access management**: web platform, server, mobile, DevOps. I came to oversee the **entire end-to-end platform** of the service, which today stably handles **over 300,000 daily accesses** without interruption. In parallel, blockchain timestamping solutions for fashion, automotive and nautical clients.

Banking · Tier 1 EU · Identity & Access Mgmt · 300k+ logins/day · React Native · JS / TS · Python · DevOps · Mobile (iOS/Android/Huawei/WP) · Blockchain Timestamping · Cybersecurity

Oct 2019 — Nov 2019 Turin · Italy

Researcher — LINK Foundation

Research activity on **Hyperledger Suite** and Elastic Search indexing for enterprise blockchain solutions.

Research · Blockchain · Hyperledger · Elastic Search

Aug 2019 — Sep 2019 Turin · Italy

Software Developer — InteK AI

Development of a **C# DLL library** to integrate the **NEO** blockchain (formerly Antshares) with the electronic electoral systems of Cyprus. Hands-on exposure to the **.NET ecosystem** and Visual Studio in a governmental context.

C# / .NET · Visual Studio · NEO Blockchain · Government · IoT

Apr 2018 — Jun 2019 Turin · Italy

Research Grant — Politecnico di Torino

Research on **databases and storage technologies for blockchain**, IPFS, distributed systems. Support for the publication of a scientific paper on distributed database types.

Research · Databases · Blockchain · IPFS

Sep 2018 — Apr 2019 Turin · Italy

M.Sc. Thesis — Politecnico di Torino · General Motors

Supporting Connected Manufacturing using Blockchain in the Automotive Domain. Architecture design, development of a working prototype, integration of three **Industry 4.0** enabling technologies: IoT, 3D printing and blockchain. Solidity smart contracts on Ethereum, MakerBot Replicator interfaced with a Raspberry Pi.

Automotive · Industry 4.0 · Ethereum / Solidity · 3D Print · IoT · Software Architecture · Traceability

Apr 2018 — Sep 2018 Turin · Italy

Software Developer — CSP

Health and environmental sustainability sectors. PHP fullstack development (SLIM micro-framework), gamification with **Unity 3D** for hospital rehabilitation projects (T.R.E.E.), IoT with MQTT.

PHP / SLIM · Unity 3D · C# / WPF · Health · IoT / MQTT · Gamification

FOCUS 2025-2026

In recent months I have moved most of my engineering work inside the agentic loop. I use **Claude** every day alongside other models, and when a tool is missing I build it. That is how the systems below were born, and today my team and I use them.

01 / MEMORY

Long-term memory for agents — Agents remember across sessions: notes classified by concept, semantic search, memory shared by several people.

02 / KNOWLEDGE

A verifiable library for LLMs — More than 120 books turned into about 40,000 linked atomic notes, each with the exact citation of its source.

03 / ORCHESTRATION

Graph-based work for people and agents — Long projects broken down into a task graph, where each node says who runs it, person or agent, and what can be picked up right now.

04 / COLLABORATION

Agent teams drawn as nodes — A visual editor where you assemble a team of agents, start it and watch it work.

05 / SKILLS

Skills and agent governance — Skills, sub-agents and hooks that teach agents how to work, with retrospectives that improve them every session.

06 / DELIVERY

Agents in everyday work — Agentic tools for the team: meeting transcription with speaker names, live call assistance, company RAG, multi-model proxy.

STACK & ENGINEERING

LLM & AI ENGINEERING

RAG Architecture — Vector search, hybrid search, graph RAG

Agent engineering — Claude Code, skills, sub-agents, hooks, automated retrospectives

Agent memory — Multilingual embeddings, concept ontologies, multi-user sync, OKF knowledge graphs

Agentic Systems — Tool use, multi-agent orchestration, planning loops

Model Context Protocol — Server, client, integrations in production

Multi-LLM Abstraction — LiteLLM, Rig-core, model routing

Embedding Pipelines — Semantic indexing, chunking, citation tracking

BACKEND · API · DATA

API Design REST/OpenAPI-first — Automatic client and schema generation

Backend frameworks — FastAPI, Axum (Tokio), Express

Async Task Processing — Celery, Tokio, background queues, event-driven microservices

Relational persistence — PostgreSQL, SQLAlchemy, query optimisation, SQLite (edge)

Specialised persistence — Vector DB (Qdrant), Graph DB (Neo4j), full-text (Elasticsearch), Redis

FRONTEND · DEVOPS · TESTING

Frontend Stack — React/Next.js, TypeScript strict, Tailwind, Shadcn

Modern tooling — Bun runtime, monorepo (Bun + Cargo workspaces), uv, Cargo, Go modules

Container & Deploy — Docker/Compose, Vercel, Raspberry Pi edge deployment

Quality Gates — MyPy, TypeScript strict, Ruff, Clippy, ESLint, pre-commit hooks

Testing — Pytest + pytest-asyncio, Playwright E2E, Cargo test, pragmatic TDD

EDUCATION · REGISTRY · LANGUAGES

EDUCATION

M.Sc. in Computer Engineering — Software specialist · Politecnico di Torino April 2019

B.Sc. in Computer Engineering — Politecnico di Torino October 2015

High School Diploma — Scientific Lyceum — Liceo Edoardo Amaldi · Novi Ligure July 2010

QUALIFICATIONS

Chartered Information Engineer — Registered with the Order of Engineers of Alessandria · Section A April 2023

Prof. Fabrizio Lamberti and Prof. Valentina Gatteschi · Politecnico di Torino · contacts on request

LANGUAGES

Italian — Native

English — Professional · B2/C1

French — Elementary · A2

INSTITUTIONAL ACTIVITY & CULTURAL INTERESTS

Founding member — Aerospace Commission

`Order of Engineers of the Province of Alessandria`

Since 2025 I have been part of an institutional initiative aimed at stimulating the **conversion of the Alessandria industrial fabric** from automotive to **aerospace and defence**, in response to the automotive crisis and current geopolitical instability.

Frontier research in biomedicine

`Personal project · open science`

I run an independent research programme on **gene therapy** for an inherited condition. It includes a state-of-the-art review published open access after peer review, an original computational model and a clinical analysis track.

Geopolitics, strategic studies and dual-use technology

`Systematic reading · personal interest`

For several years I have devoted constant attention to **geopolitical scenarios**, international security dynamics, Defense industrial doctrines, technological competition between blocs and **dual-use** policies.